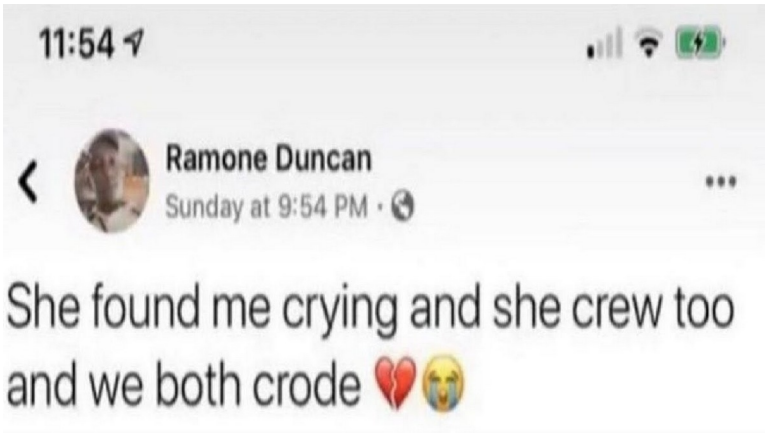
# More modeling: The case of inflectional morphology and specific language impairment

# Quick review

**Morphology is “generative”**

For example, at some point there was no noun “generative,” just the verb “to generate.” But English has a process for turning verbs into nouns. That’s *derivational morphology.*

Today we’re going to focus on *inflectional morphology.* In English this pretty much means the plural and the past tense.

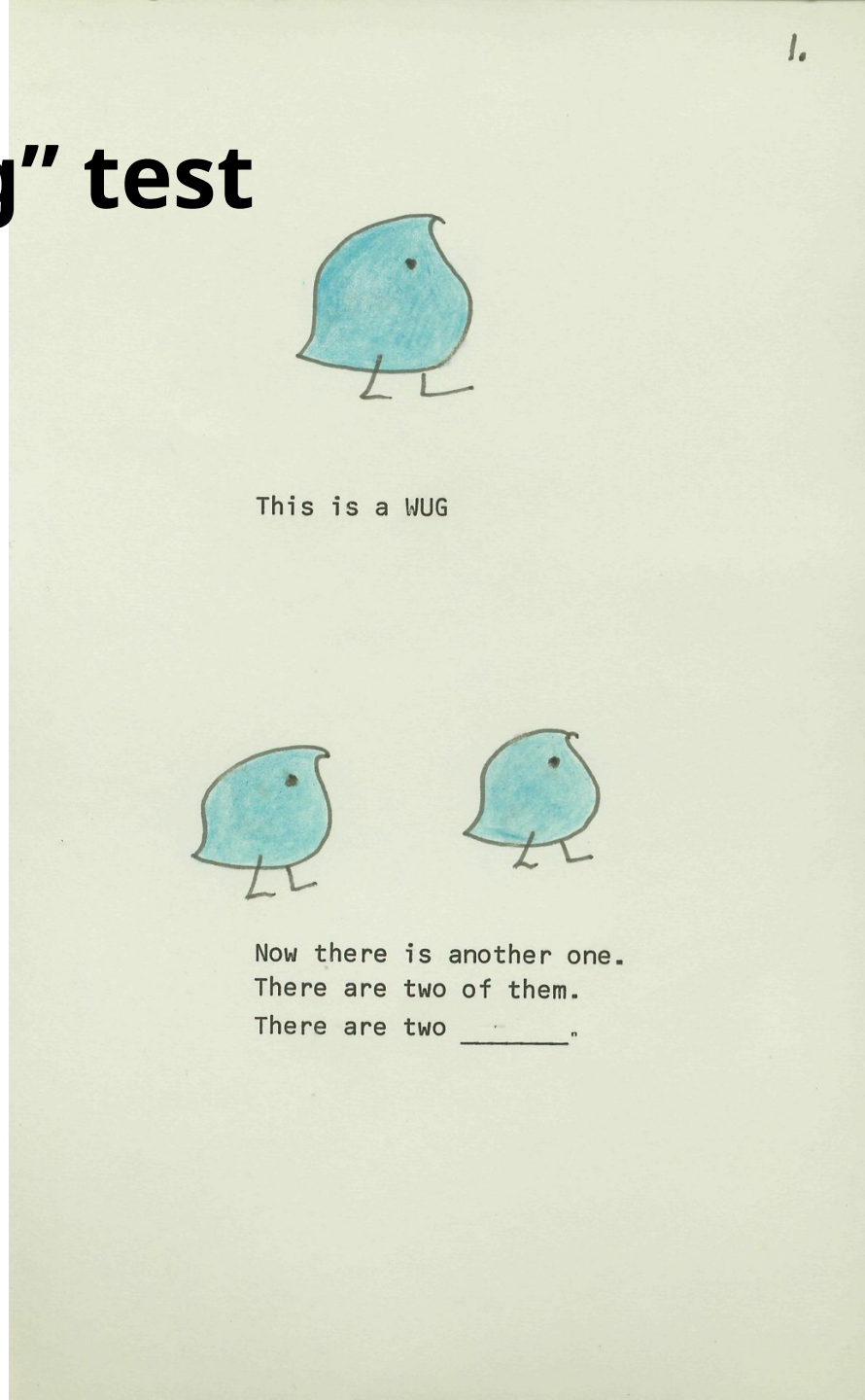


# The “wug” test

Classic study by Jean Berko Gleason (1958).

Demonstrates how readily people generalize the past tense to new words.

Children do this too (thus the cute drawings).



# The English past tense has “regular” and “irregular” forms.

**Regulars**

show-showed, bake-baked, wade-waded

blick-blicked, trog-trogged, wumble-wumbled

**Irregulars**

go-went, take-took, eat-ate, sleep-slept

*This should remind you right away of the last few lectures…*

# Words and Rules

Pinker & colleagues

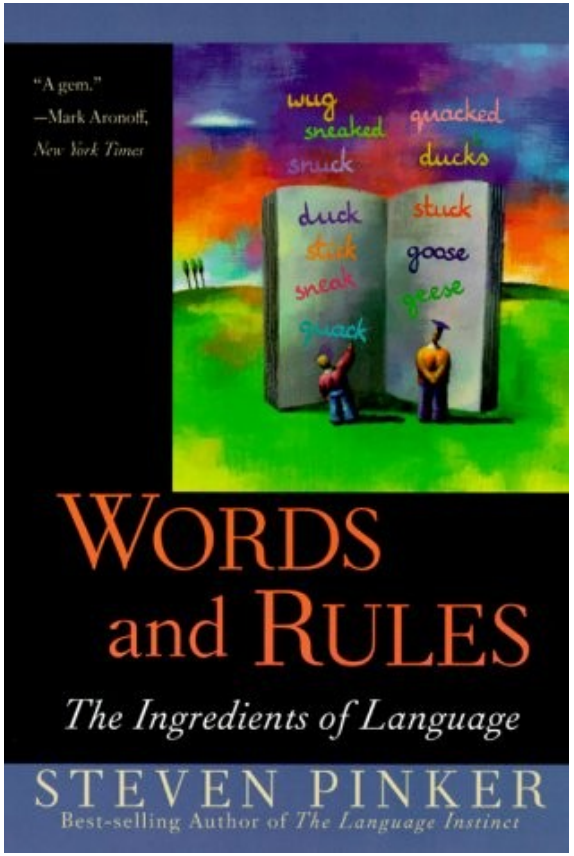
‘words and rules’ are separate processes, representational formats, etc.

**Regulars**:

symbolic rules (e.g., “add –ed”)

**Irregulars:**

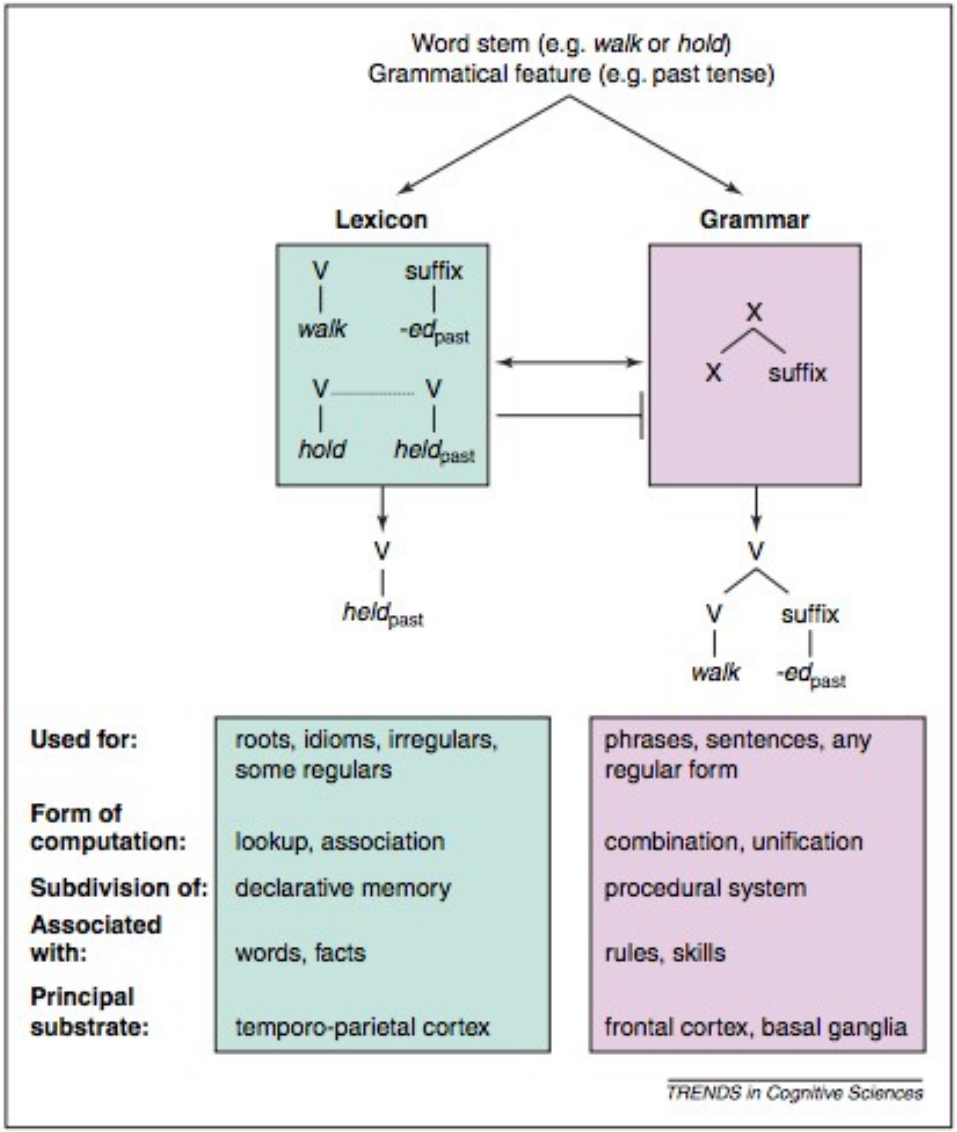
lexicalized within “associative network”



# A “dual route” model for the past tense

How is this similar to the dual-route model we were just considering for reading?

How is it different?




*Pinker & Ullman (2002,*

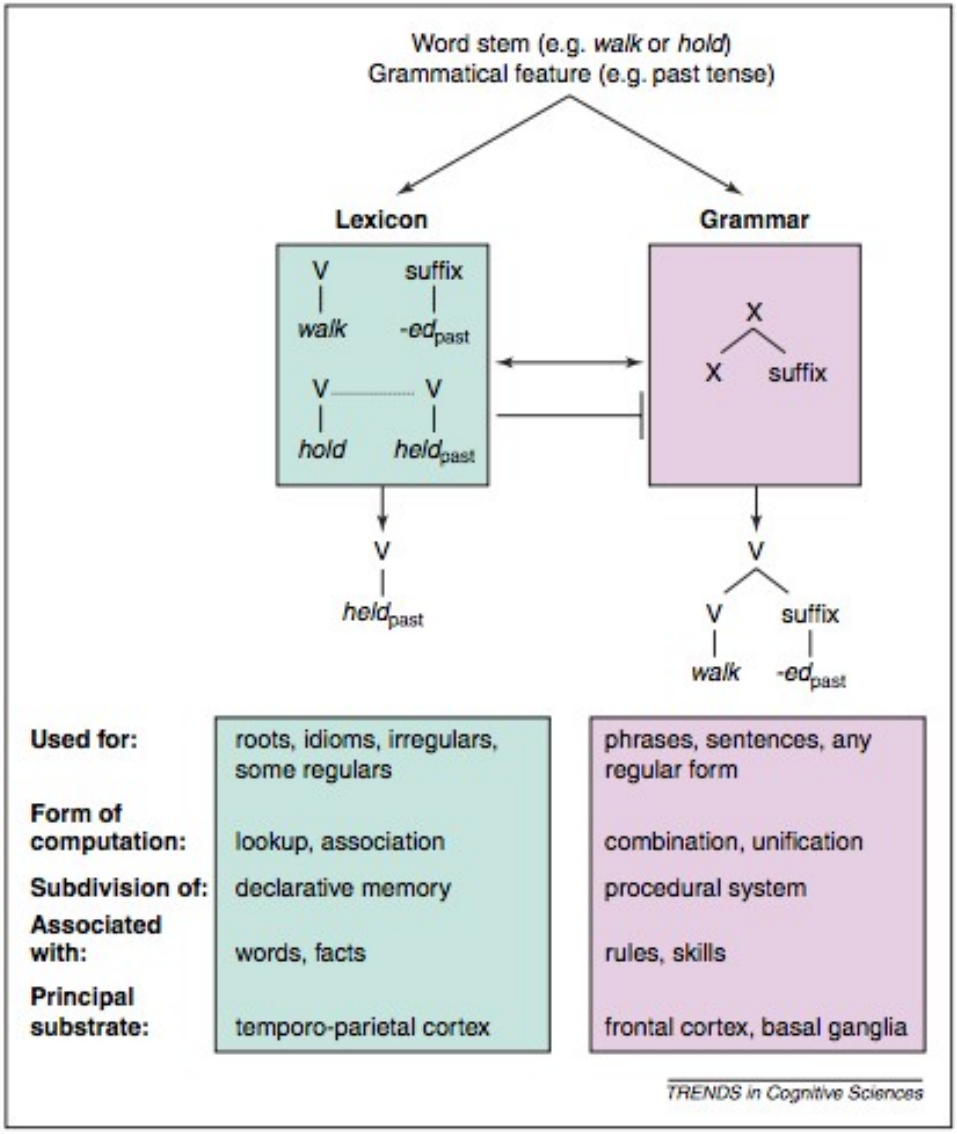
# Similar architecture, different origins

Remember the dual route model was developed based on neuropsychological double-dissociations.

Pinker et al. are motivated by other concerns.

Generative linguistics of the time was mostly about categorical, symbolic, rule-based grammars.

These rules often have exceptions.



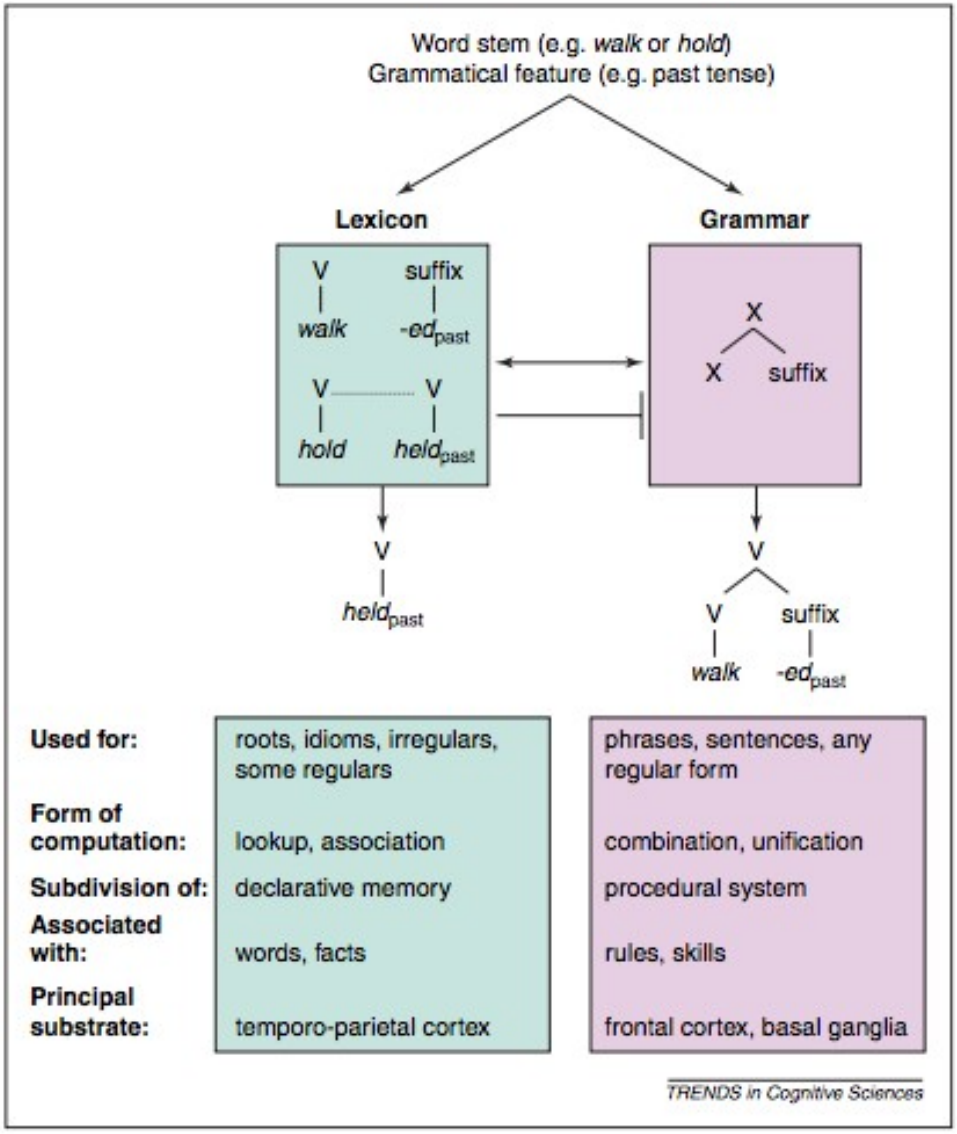
# Similar architecture, different origins

Rather than make the grammar inelegant by adding machinery to handle the exceptions, linguists would often just leave the exceptions out of the grammar.

So dual-process model proposes a way to “handle” the exceptions while keeping the grammar elegant, symbolic, categorical, etc.

This second process had all the properties the rule route did not. It was:

- word-specific (lexical)
  - sensitive to frequency of occurrence
  - sensitive to phonological



# An alternative approach: It's all associative learning!

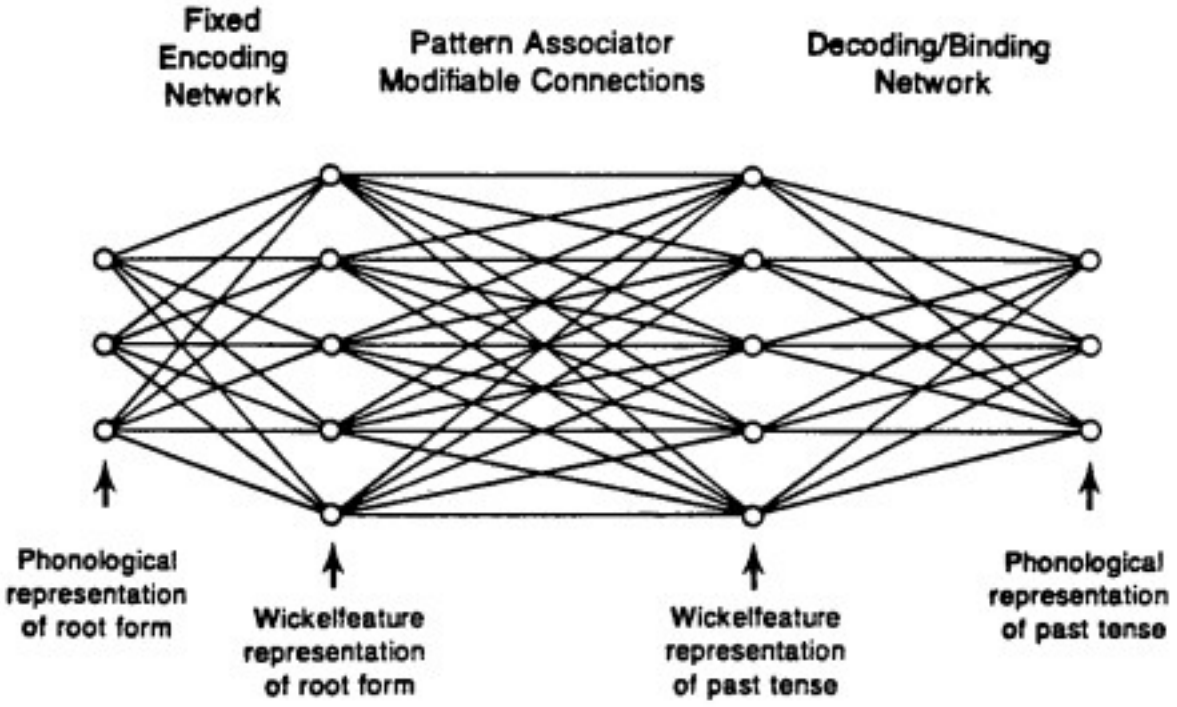



FIGURE 1. The basic structure of the model.

*Rumelhart & McClelland (1987) available, along with lots of these other papers, here: http://psych.stanford.edu/~jlm/papers/*

# Target phenomena of Rumelhart & McClelland

Because this was one of the first *learning models* to address a language process, the focus was on capturing developmental phenomena.

Specifically: “U-shaped” learning:

- some frequent irregular verbs are learned quite early, e.g., *run-ran*

- then, spontaneously, children begin producing “regularizations:” *runned.*

- eventually, they start producing correct forms again.

Learning is “U-shaped” because children often get worse before they get better, so performance goes down before it goes back up.

# Why is U-shaped learning important to the “words and rules” approach?

It’s understood in terms of distinct stages/processes:

- early correct forms are just imitations, don’t depend on the same mechanisms as adult processing.

- when the rule is initially learned, it is applied to all cases. We can think of the rule as the “best hypothesis” for all items generally.

- over time, exceptions are learned associatively, one-by-one and incorporated into lexical knowledge. Further, the “blocking process” develops.

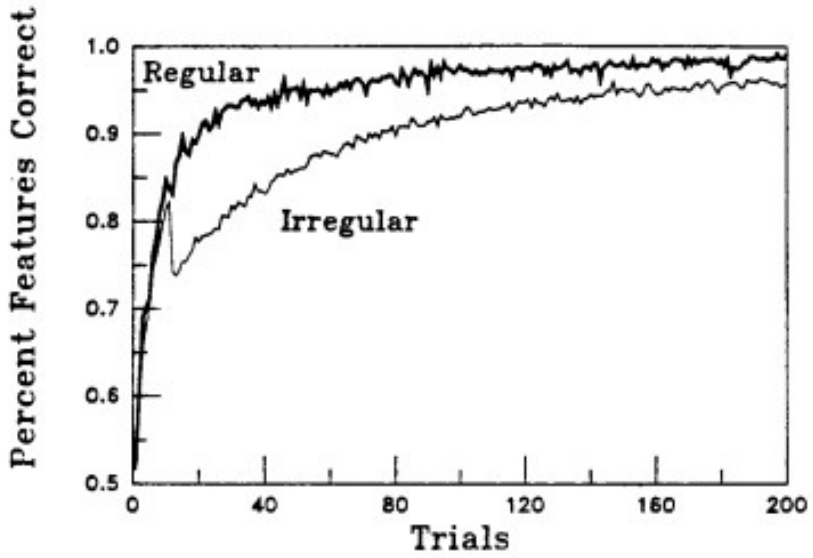



FIGURE 4. The percentage of correct features for regular and irregular high-frequency verbs as a function of trials.

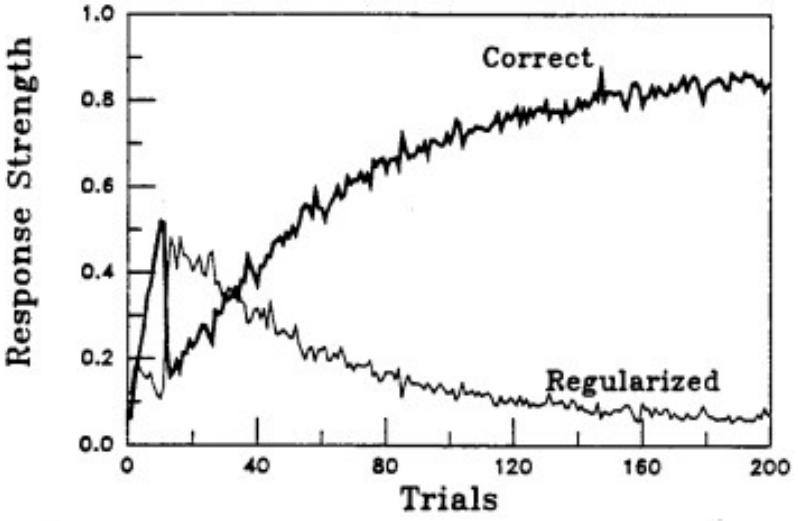



FIGURE 5. Response strengths for the high-frequency irregular verbs. The response strengths for the correct responses are compared with those for the regularized alternatives as a function of trials.

# "U-shaped learning" in Rumelhart & McClellands's model.

- training set is based on data about input to children, which changes over time.

– Early on, irregular verbs predominate – in fact the highest-frequency verbs tend to have irregular past-tense forms: GIVE, TAKE, HAVE, GO, etc.

– As the vocabulary expands to include more regular forms, the model reorganizes to capture those regularities.

–- this interferes with some of the early-learned forms

TABLE 18

SYSTEM RESPONSES TO UNFAMILIAR LOW-FREQUENCY REGULAR VERBS

| Verb Type | Presented Word | Phonetic Input | Phonetic Response | English Rendition | Response Rendition |
|---|---|---|---|---|---|
| End in *t/d* | guard | /gɑrd/ | /gɑrd/ | (guard) | 0.29 |
| | | | /gɑrd^d/ | (guarded) | 0.26 |
| | kid | /kid/ | /kid/ | (kid) | 0.39 |
| | | | /kid^d/ | (kidded) | 0.24 |
| | mate | /mAt/ | /mAt^d/ | (mated) | 0.43 |
| | | | /mAd^d/ | (maded) | 0.23 |
| | squat | /skw*t/ | /skw*t^d/ | (squated) | 0.27 |
| | | | /skw*t/ | (squat) | 0.22 |
| | | | /skw*kt/ | (squawked) | 0.21 |
| End in unvoiced consonant | carp | /kɑrp/ | /kɑrpt/ | (carped) | 0.28 |
| | | | /kɑpt^d/ | (carpted) | 0.21 |
| | drip | /drip/ | /dript^d/ | (dripted) | 0.28 |
| | | | /dript/ | (dripped) | 0.22 |
| | map | /map/ | /mapt^d/ | (mapted) | 0.24 |
| | | | /mapt/ | (mapped) | 0.22 |
| | shape | /sAp/ | /sApt/ | (shaped) | 0.43 |
| | | | /sipt/ | (shipped) | 0.27 |
| | sip | /sip/ | /sipt/ | (sipped) | 0.42 |
| | | | /sept/ | (sepped) | 0.28 |
| | slip | /slip/ | /slept/ | (slept) | 0.40 |
| | smoke | /smOk/ | /smOkt^d/ | (smokted) | 0.29 |
| | | | /smOk/ | (smoke) | 0.22 |
| | snap | /snap/ | /snapt^d/ | (snapted) | 0.40 |
| | step | /step/ | /stept^d/ | (stepted) | 0.59 |
| | type | /tip/ | /tipt^d/ | (typted) | 0.33 |
| End in voiced consonant or vowel | brown | /brwn/ | /brwnd/ | (browned) | 0.46 |
| | | | /br*nd/ | (brawned) | 0.39 |
| | hug | /h^g/ | /h^g/ | (hug) | 0.59 |
| | mail | /mA^l/ | /mA^ld/ | (mailed) | 0.38 |
| | | | /memb^ld/ | (membled) | 0.23 |
| | tour | /tUr/ | /tUrd^r/ | (toureder) | 0.31 |
| | | | /tUrd/ | (toured) | 0.25 |

Generalization testing:

The model was trained on a small number of verbs, so they could do a “wug” test with real words…

TABLE 16

PROPORTION OF WICKELFEATURES CORRECTLY GENERATED

| | |
|---|---|
| Regular | .92 |
| Irregular | .84 |
| Overall | .91 |

# But...

Finally, there were just four responses that involved the addition or modification of consonants. These were *maded* as a past tense of *mate*, *squawked* as a past tense for *squat*, *membled* as a past tense for *mail*, and *toureder* as a past tense for *tour*. It is unlikely that humans would make these errors, especially the last two, but these responses are, for the most part, near threshold. Furthermore, it seems likely that many of these responses could be filtered out if the model incorporated an auto-associative network of connections among the output units. Such a network could be used to clean up the output pattern and would probably increase the tendency of the model to avoid bizarre responses. Unfortunately, we have not yet had the chance to implement this suggestion.

2.2. *Keeping individuals from blending.* With feature-only representations, representing two things simultaneously is problematic. If early visual input is an activation pattern over feature maps, then without a serial attentional mechanism you can't tell the difference between a green circle near a red square and a red circle near a green square. In the Rumelhart & McClelland model, various subregularities and the regular rule all compete to activate Wickelfeatures for the past form: When *sing* is input, you could get features for *sing*, *sang*, *sung*, *singed*, *sanged*, etc., all superimposed. The model isn't putting out any single word at all, just an unordered collection of features, many of them contradictory. Thus it couldn't avoid blending what should be distinct competing outputs into bizarre hybrids such as *toureder* as the past of *tour*, or *membled* for *mail*.

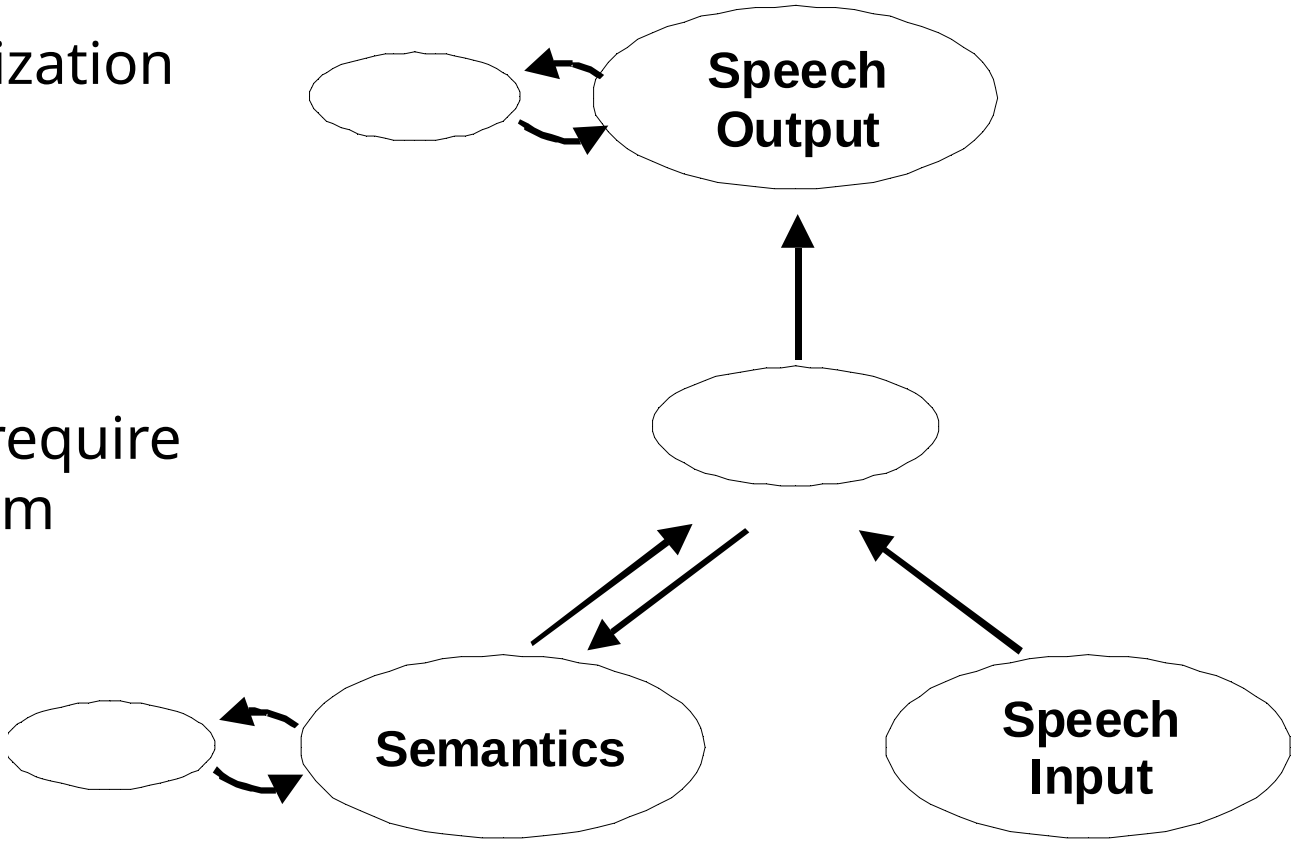
*Prince & Pinker (1988, BBS)*

# A new(er) model: Joanisse & Seidenberg (1999)

Theoretical premise:

- “rule-like” generalization is supported by phonological representations

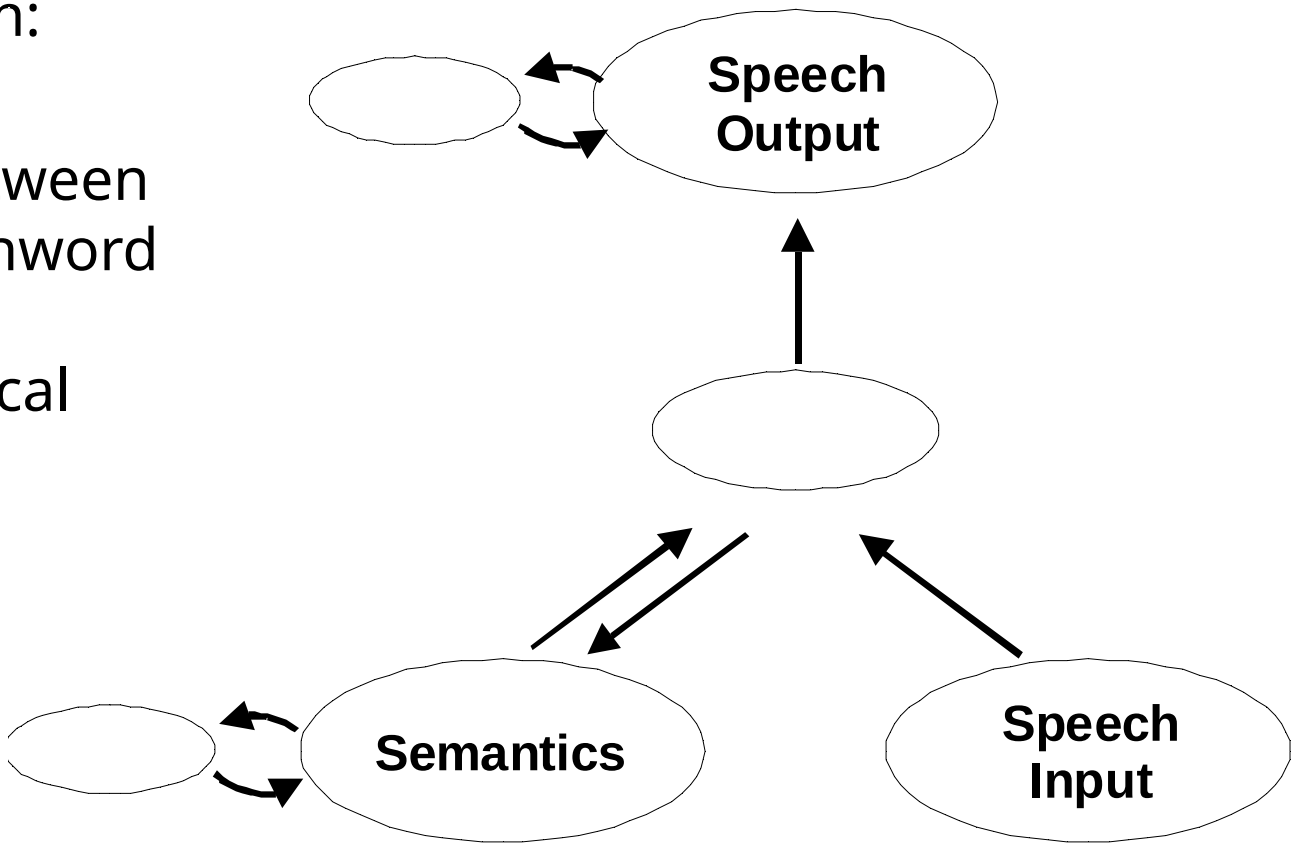
- “irregular” forms require greater support from semantics



# A new(er) model: Joanisse & Seidenberg (1999)

Target phenomenon:

- dissociation between irregular and nonword past tense in neuropsychological data.



# Neuropsychological data

## Broca's Aphasia

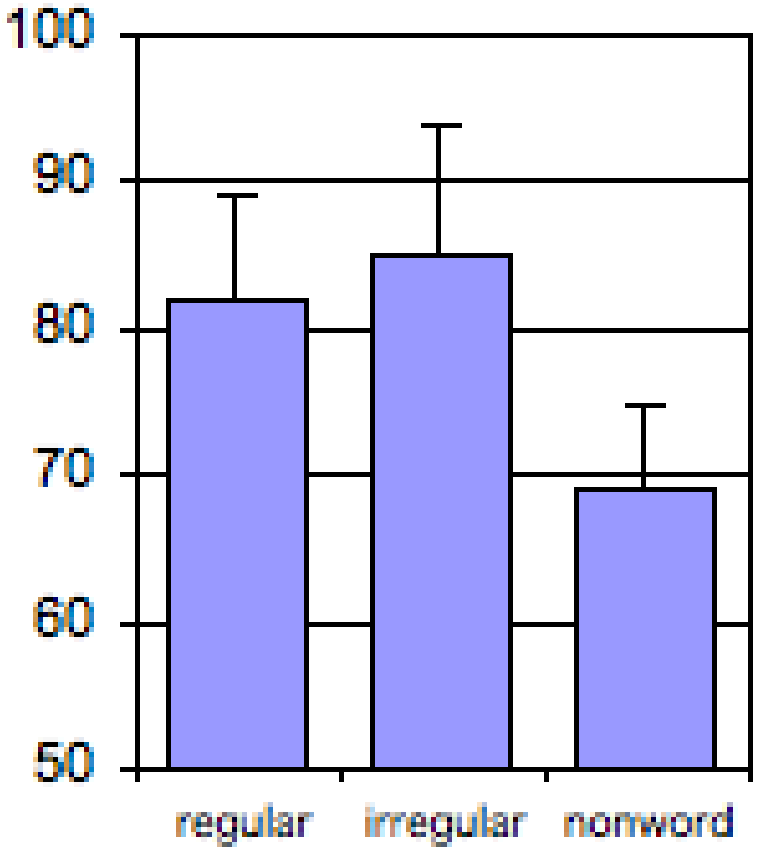
Injury to left anterior brain regions

Poor speech production

Problems with regular morphology, esp. nonwords

*wug* *wugged,*

*blick* *blicked*

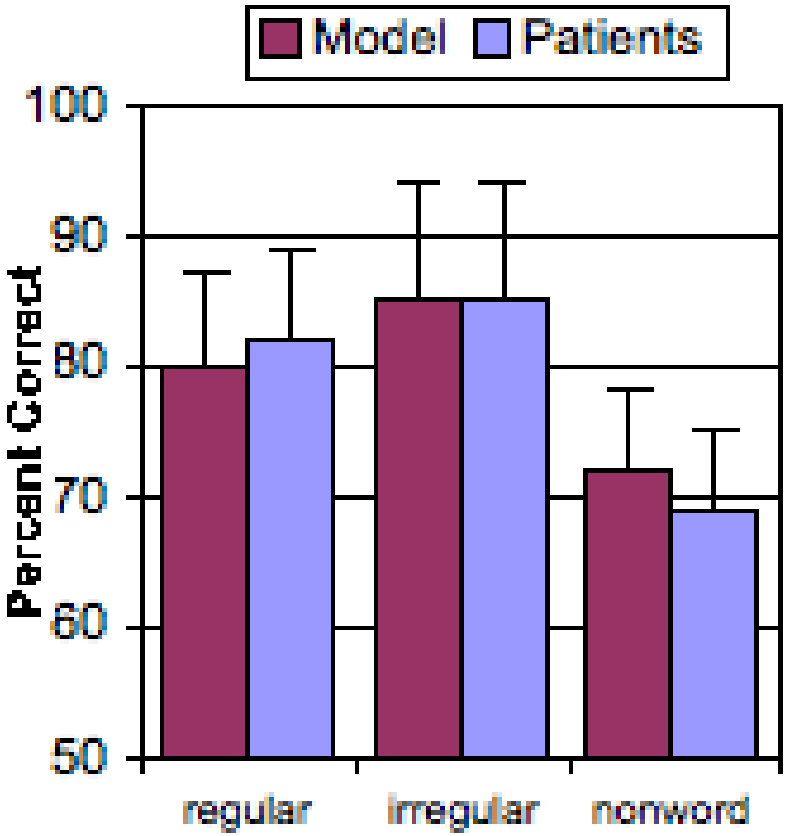


# Phonological Impairment in the Model

Reduced ability to compute past tense forms

Nonwords worse than regulars and irregulars

Regulars numerically worse than irregulars

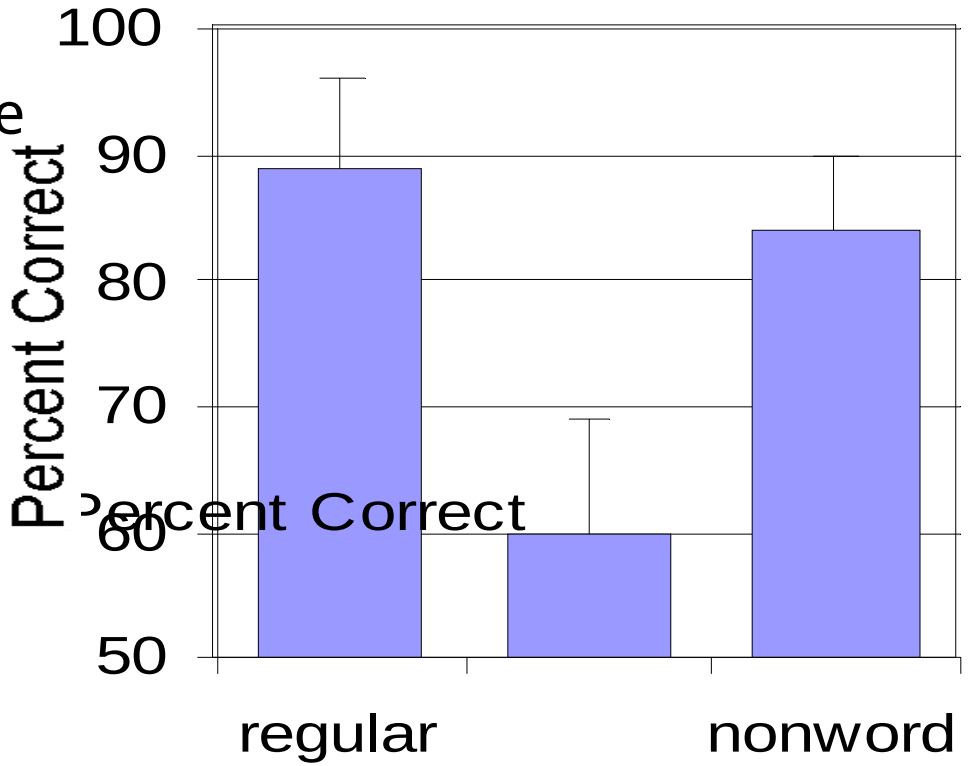


# Neuropsychological data: Problems with irregulars

Wernicke's and Alzheimer's patients

Anomic patients with injury to inferior/middle temporal brain regions

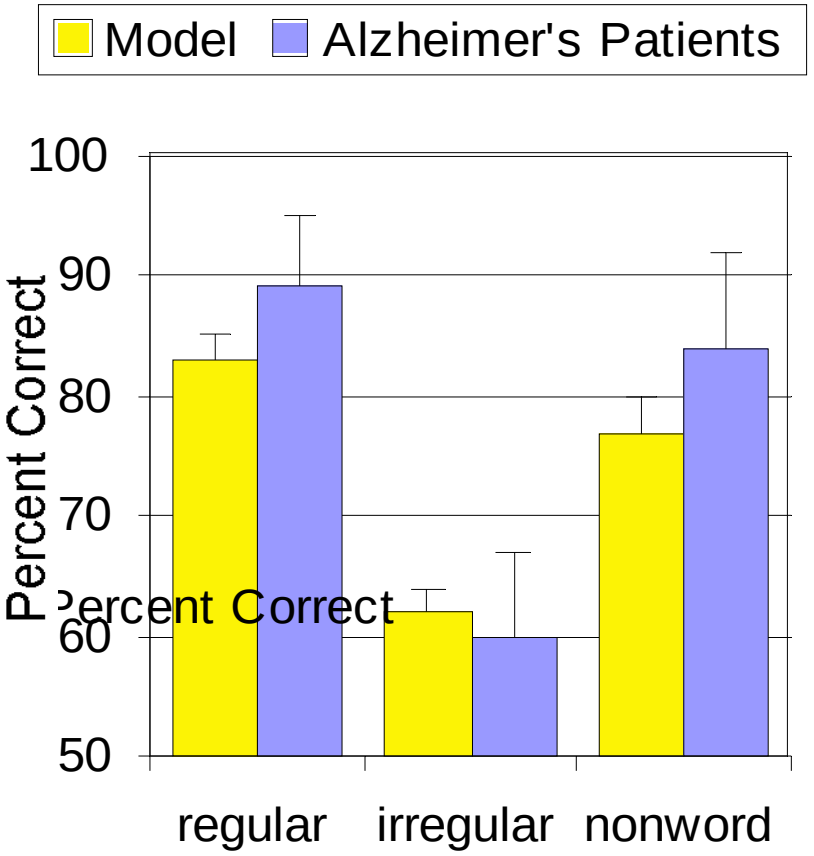
Poor on irregulars (middle bar)



# Effects of Semantic Impairment

Irregulars worse than regulars and nonwords

Nonwords a bit worse than regulars



# Other issues with the double dissociation data:

Broca's patients:

Differences between regulars and irregulars are reduced when items are adequately controlled for phonology

(Bird, Patterson, Lambon-Ralph, McClelland & Seidenberg, 2003)

Semantic dementia patients:

Degree of impairment with irregulars correlates strongly with severity of semantic deficit

(Patterson, Lambon-Ralph, Hodges & McClelland, 2001)

# Compare this to the “triangle model” account of dyslexia:

**Phonological representations support generalization.**

generalization is a result of statistical learning about patterns in the mapping from present to past

impaired phonological representations impact these generalizations

**Semantic representations support idiosyncratic forms.**

forms that deviate from the overwhelmingly regular “add –ed” pattern need support from semantics

some “irregular” forms can take advantage of the sub-patterns, and are likely to be “spared”

# Specific Language Impairment

- Impaired language development
- “Normal” performance/nonverbal IQ
- Affected grammar
  - morphology
  - syntax
- But also:
  - phonology
  - reading
  - working memory
  - speech perception

# Past Tense and SLI

Past tense deficits: *sine qua non* of SLI

Earlier claims suggested their deficit was specific to rules

Gopnik; van der Lely; Pinker; Ullman; Clahsen

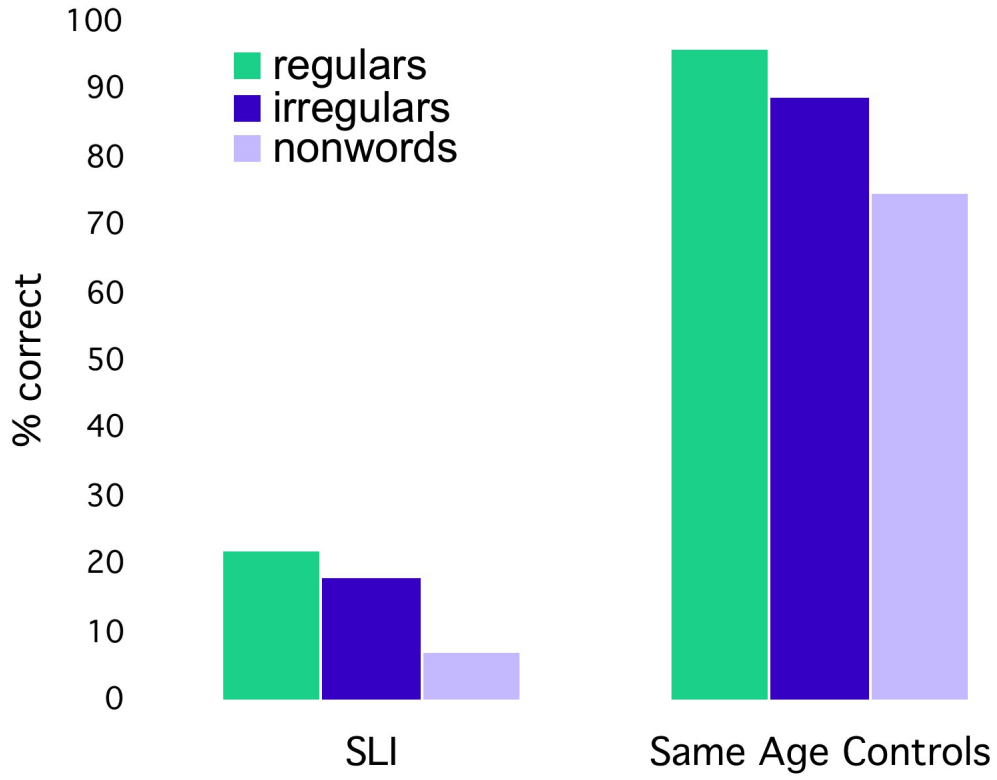
SLI could represent evidence that language acquisition requires domain-specific mechanisms

*A deficit in learning rules*

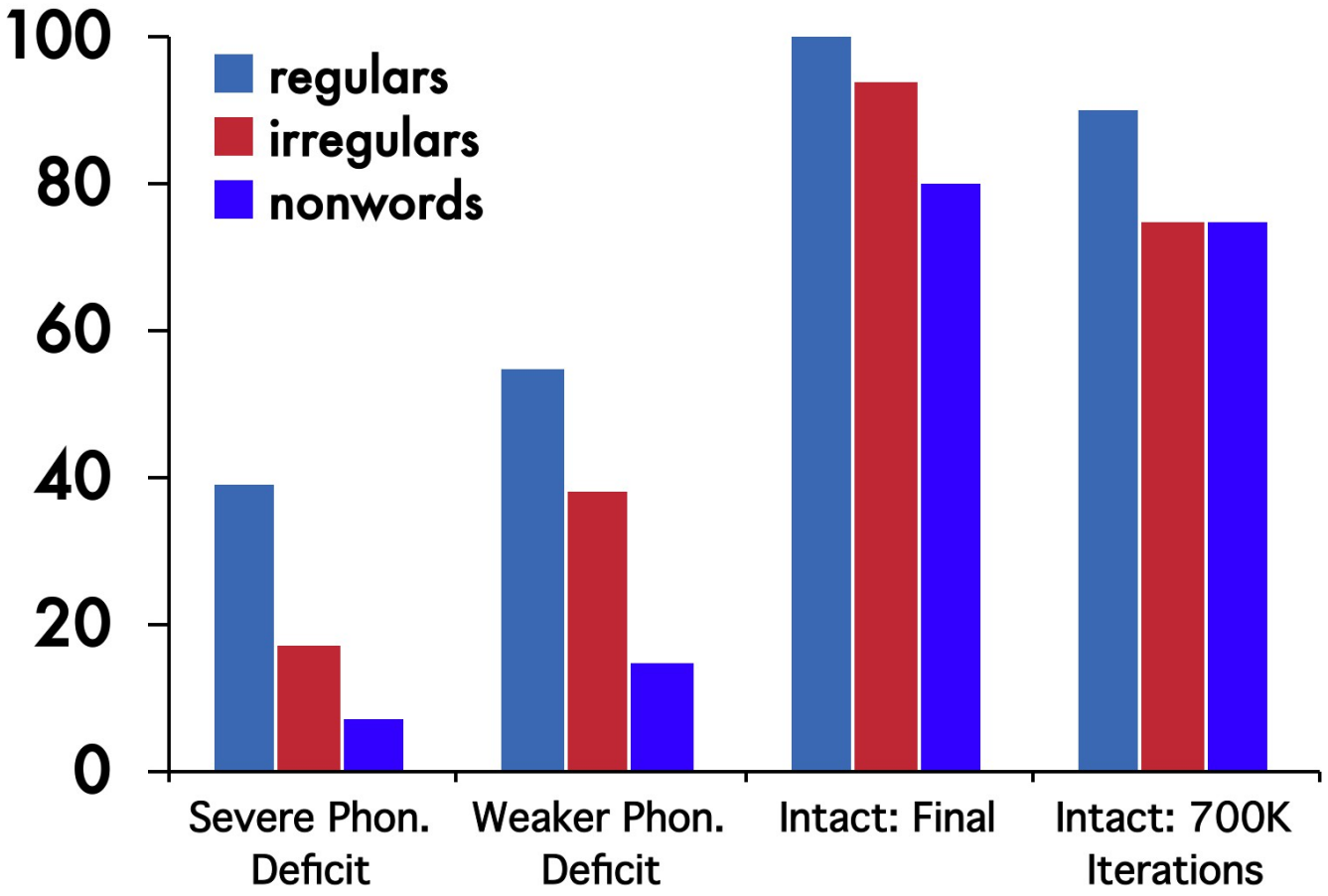
a problem with regulars and nonwords, but *not* irregulars

# Behavioral Data

van der Lely & Ullman (2001)



# Model Results



# Summary

The model illustrates how phonology affects development of all forms

…and why it’s especially important to nonwords

# Note similarities to “triangle” model of reading.

There are no “special rules” for the past tense, just mappings between phonology and semantics.

There is no separate lexicon that stores the exceptions.

Indeed, the concept of “rule” and “exception” only approximately describes the system.

*Association* between “irregular” forms and semantic processing.

*Association* between “wug test” performance and phonological processing.

For next lecture:

Read Borodisky (2003) on Linguistic Relativity